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TRAINING DATA GENERATION APPARATUS, METHOD, AND NON-TRANSITORY COMPUTER READABLE MEDIUM

Abstract

According to one embodiment, a training data generation apparatus includes a processor. The processor acquires operation data related to an operation state of a device in a predetermined period. The processor divides the operation data into at least temporary training data and temporary test data. The processor detects an anomaly value from the temporary test data based on the temporary training data. The processor generates training data from the operation data by excluding an anomaly period in which the anomaly value is detected from the predetermined period.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-036095, filed Mar. 8, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] Embodiments described herein relate generally to a training data generation apparatus, method, and non-transitory computer readable medium.

BACKGROUND

[0003] An industrial system (examples: infrastructure facility, manufacturing facility) includes a plurality of devices, and a process of each device is controlled. Stopping the industrial system incurs social and economic losses. In order to keep an industrial system healthy, it is necessary to monitor an operation state of the industrial system.

[0004] For example, an operation state of the industrial system is monitored by a machine training model. The machine training model uses, as training data, operation data (normal data) in a period (normal period) in which the device is in a normal operation state. In a case of being trained on this training data, the machine training model can detect an anomaly value from arbitrary operation data of the device.

[0005] Conversely, the machine training model may use operation data (anomaly data) for a period (anomaly period) in which the device is in an anomaly operation state as training data. In a case where the machine training model is trained on the training data, it is difficult for the machine training model to detect an anomaly value from arbitrary operation data of the device. Therefore, it is required to appropriately generate the training data so that the training data includes normal data and does not include anomaly data.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0006] FIG. 1 is a functional configuration diagram of a training data generation apparatus according to a first embodiment.

[0007] FIG. 2 is an operation flowchart of the training data generation apparatus according to the first embodiment.

[0008] FIG. 3 is an illustrative diagram illustrating training data generation processing according to the first embodiment.

[0009] FIG. 4 is an illustrative diagram illustrating division processing of operation data according to a first modification of the first embodiment.

[0010] FIG. 5 is an illustrative diagram illustrating division processing of operation data according to a second modification of the first embodiment.

[0011] FIG. 6 is an illustrative diagram illustrating division processing of operation data according to a third modification of the first embodiment.

[0012] FIG. 7 is an illustrative diagram illustrating anomaly value detection processing according to the third modification of the first embodiment.

[0013] FIG. 8 is an illustrative diagram illustrating division processing of operation data according

to a fourth modification of the first embodiment.

[0014] FIG. **9** is a first illustrative diagram illustrating anomaly value detection processing according to a fourth modification of the first embodiment.

[0015] FIG. **10** is a second illustrative diagram illustrating anomaly value detection processing according to the fourth modification of the first embodiment.

[0016] FIG. **11** is a functional configuration diagram of a training data generation apparatus according to a second embodiment.

[0017] FIG. **12** is an operation flowchart of the training data generation apparatus according to the second embodiment.

[0018] FIG. **13** is a frequency distribution diagram of temporary training data according to a second modification of the second embodiment.

[0019] FIG. **14** is a frequency distribution diagram according to a third modification of the second embodiment.

[0020] FIG. **15** is an illustrative diagram illustrating anomaly value detection processing according to a fourth modification of the second embodiment.

[0021] FIG. **16** is a functional configuration diagram of a training data generation apparatus according to a third embodiment.

[0022] FIG. **17** is an operation flowchart of the training data generation apparatus according to the third embodiment.

[0023] FIG. **18** is an illustrative diagram illustrating anomaly value detection processing according to the third embodiment.

[0024] FIG. **19** is a frequency distribution diagram of operation data and reference data according to the third embodiment.

[0025] FIG. **20** is a hardware configuration diagram of the training data generation apparatus according to each embodiment.

DETAILED DESCRIPTION

[0026] In general, according to one embodiment, a training data generation apparatus includes a processor. The processor acquires operation data related to an operation state of a device in a predetermined period. The processor divides the operation data into at least temporary training data and temporary test data. The processor detects an anomaly value from the temporary test data based on the temporary training data. The processor generates training data from the operation data by excluding an anomaly period in which the anomaly value is detected from the predetermined period.

[0027] Hereinafter, embodiments will be described with reference to the drawings. In the following embodiments, portions denoted by the same reference numerals perform similar operations, and redundant description will be omitted as appropriate.

First Embodiment

[0028] FIG. **1** is a functional configuration diagram of a training data generation apparatus **1** according to a first embodiment. The training data generation apparatus **1** is an apparatus that generates training data XD from operation data OD. The training data generation apparatus **1** includes an acquisition unit **11**, a division unit **12**, a detection unit **13**, and a generation unit **14**.

[0029] The acquisition unit **11** acquires various data. The acquisition unit **11** acquires the operation data OD regarding an operation state of the device in a predetermined period. The acquisition unit **11** acquires the operation data OD from the outside of the training data generation apparatus **1**. The acquisition unit **11** transmits the acquired operation data OD to the division unit **12** and the generation unit **14**. The operation data OD is also referred to as “candidate data”.

[0030] The division unit **12** divides various data. The division unit **12** divides the operation data OD into at least temporary training data RD and temporary test data SD. The division unit **12** transmits the divided temporary training data RD and temporary test data SD to the detection unit **13**.

[0031] The detection unit **13** detects various data. The detection unit **13** detects an anomaly value AV from the temporary test data SD based on the temporary training data RD. The detection unit **13** transmits the detected anomaly value AV to the generation unit **14**. The anomaly value AV is also referred to as an “anomaly candidate”. The anomaly value AV may include an anomaly sign.

[0032] The generation unit **14** generates various data. The generation unit **14** generates the training data XD from the operation data OD by excluding a period (anomaly period) in which the anomaly value AV is detected from a predetermined period of the operation data OD. The generation unit **14** outputs the generated training data XD to the outside of the training data generation apparatus **1**.

[0033] FIG. **2** is an operation flowchart of the training data generation apparatus **1** according to the first embodiment. The training data generation apparatus **1** sequentially executes each processing according to steps **S11** to **S14**.

[0034] (Step **S11**) First, the training data generation apparatus **1** acquires the operation data OD through the acquisition unit **11**. The operation data OD is time-series data related to an operation state of a predetermined device in an industrial system. The operation data OD may be time-series data related to each of a plurality of process variables indicating an operation state of a predetermined device. Each process variable may be a measured value (examples: temperature, pressure, speed, acceleration, current, voltage) from each sensor installed in a predetermined device.

[0035] In a case where a predetermined device is affected by the periodic variation, the period of the operation data OD related to the predetermined device may be set to one cycle or more. For example, in a case where the predetermined device is affected by the seasonal variation, the period of the operation data OD is set to one year or more. The operation data OD may include values sampled from each sensor at any time interval (e.g., 1 minute).

[0036] (Step **S12**) Next, the training data generation apparatus **1** divides the operation data OD through the division unit **12**. The division unit **12** divides the operation data OD acquired in step **S11** into the temporary training data RD and the temporary test data SD. The period of the temporary training data RD does not overlap the period of the temporary test data SD. The period of the temporary training data RD may be longer than the period of the temporary test data SD.

[0037] In particular, in a case where the predetermined device is affected by the seasonal variation, the period of the temporary test data SD related to the predetermined device may be set to be less than the length of each season of spring, summer, and autumn (for example, less than three months in Japan). In a case where the period of the temporary test data SD is set to be equal to or longer than the length of each season, it can be erroneously detected that the seasonal variation such as the air temperature and the water temperature is the anomaly value AV.

[0038] (Step **S13**) Subsequently, the training data generation apparatus **1** detects the anomaly value AV through the detection unit **13**. The detection unit **13** uses the temporary training data RD and the temporary test data SD divided in step **S12**. The detection unit **13** assumes that the temporary training data RD is data in a normal state (normal data), and detects the anomaly value AV from the temporary test data SD.

[0039] The detection unit **13** may determine whether or not the value of the temporary test data SD is included in a value range between the maximum value and the minimum value related to the temporary training data RD. The detection unit **13** detects the value of the temporary test data SD not included in the value range as the anomaly value AV. The detection unit **13** may detect a period (anomaly period) in which anomaly value AV is detected in temporary test data SD.

[0040] (Step **S14**) Finally, the training data generation apparatus **1** generates the training data XD through the generation unit **14**. The generation unit **14** uses the anomaly value AV (or anomaly period) detected in step **S13**. The generation unit **14** generates the training data XD from the operation data OD by excluding the period (anomaly period) in which the anomaly value AV is detected from the operation data OD.

[0041] FIG. **3** is an illustrative diagram illustrating generation processing of the training data XD

according to the first embodiment. FIG. 3 sequentially illustrates each processing according to steps S11 to S14 in FIG. 2. A period of each data in each processing is indicated by bands B1A to B4A. For convenience of description, the period is expressed as [start time, end time] or (start time, end time). In a closed section represented by [A, B], the start time and the end time are included in the section. In an open section represented by (A, B), the start time or the end time is not included in the section.

[0042] As illustrated in a band B1A of a first stage, the acquisition unit 11 acquires the operation data OD (see step S11). The period of the operation data OD is [t1, t4].

[0043] As illustrated in a band B2A of a second stage, the division unit 12 divides the operation data OD into two pieces of temporary training data RD1 and RD2 and one piece of temporary test data SD (see step S12). The period of the temporary training data RD1 is [t1, t2], and the period of the temporary training data RD2 is [t3, t4]. The period of the temporary test data SD is (t2, t3). There is no blank period G between the temporary training data RD1 or RD2 and the temporary test data SD.

[0044] As illustrated in a band B3A of a third stage, the detection unit 13 detects the anomaly value AV from the temporary test data SD based on the temporary training data RD1 and RD2 (see step S13). The anomaly period in which the anomaly value AV is detected is [tP1, tP2].

[0045] As illustrated in a band B4A of a fourth stage, the generation unit 14 generates two pieces of training data XD1 and XD2 by excluding the anomaly period [tP1, tP2] in which the anomaly value AV is detected from the period [t1, t4] of the operation data OD (see step S14). The period of the training data XD1 is [t1, tP1], and the period of the training data XD2 is (tP2, t4].

[0046] According to the first embodiment, the training data generation apparatus 1 detects the anomaly value AV from the period of the temporary test data SD. The training data generation apparatus 1 generates the training data XD by excluding the anomaly period in which the anomaly value AV is detected from the period of the operation data OD. Therefore, the training data generation apparatus 1 can appropriately generate the training data XD such that the training data XD includes normal data and does not include anomaly data. In a case of being trained on the training data XD, the machine training model can detect an anomaly value AV from any operation data OD of the device. That is, the training data generation apparatus 1 can support anomaly detection by the machine training model.

First Modification of First Embodiment

[0047] FIG. 4 is an illustrative diagram illustrating division processing of the operation data OD according to a first modification of the first embodiment. FIG. 4 sequentially illustrates each processing according to steps S11 and S12 in FIG. 2. A period of each data in each processing is indicated by bands B1B and B2B.

[0048] As illustrated in a band B1B of a first stage, the acquisition unit 11 acquires the operation data OD (see step S11). The period of the operation data OD is [t1, t6].

[0049] As illustrated in the band B2B of a second stage, the division unit 12 divides the operation data OD into two pieces of temporary training data RD1 and RD2 and one piece of temporary test data SD (see step S12). The period of the temporary training data RD1 is [t1, t2], and the period of the temporary training data RD2 is [t5, t6]. The period of the temporary test data SD is [t3, t4].

[0050] A blank period G1(t2, t3) exists between the temporary training data RD1 and the temporary test data SD. A blank period G2(t4, t5) exists between the temporary training data RD2 and the temporary test data SD. The two blank periods G1 and G2 do not belong to the periods of the two pieces of temporary training data RD1 and RD2, and do not belong to the period of the temporary test data SD.

[0051] It is assumed that the anomaly value AV exists in at least one of the two blank periods G1 and G2 and the temporary test data SD. Further, it is assumed that there is no anomaly value AV in the temporary training data RD1 and RD2. In this case, the detection unit 13 can detect the anomaly value AV from the temporary test data SD based on the temporary training data RD1 and

RD2 not including the anomaly value AV. On the other hand, it is assumed that there is no blank period G between the temporary training data RD and the temporary test data SD, the blank period G1 is included in the temporary training data RD1, and the blank period G2 is included in the temporary training data RD2. In this case, since the temporary training data RD includes the anomaly value AV, the anomaly value AV is not detected from the temporary test data SD. [0052] According to the first modification of the first embodiment, the training data generation apparatus 1 divides the operation data OD such that the blank period G exists between the temporary training data RD and the temporary test data SD. Therefore, even in a case where the anomaly value AV is included in the blank period G adjacent to the period of the temporary test data SD, the training data generation apparatus 1 can detect the anomaly value AV from the temporary test data SD.

Second Modification of First Embodiment

[0053] FIG. 5 is an illustrative diagram illustrating division processing of the operation data OD according to a second modification of the first embodiment. FIG. 5 sequentially illustrates each processing according to steps S11 and S12 in FIG. 2. A period of each data in each processing is indicated by bands B1C and B2C.

[0054] As illustrated in a band B1C of a first stage, the acquisition unit 11 acquires the operation data OD (see step S11). The period of the operation data OD is [t1, t10].

[0055] As illustrated in the band B2C of a second stage, the division unit 12 divides the operation data OD into three pieces of temporary training data RD1 to RD3 and two piece of temporary test data SD1 and SD2 (see step S12). The period of the temporary training data RD1 is [t1, t2], the period of the temporary training data RD2 is [t5, t6], and the period of the temporary training data RD3 is [t9, t10]. The period of the temporary test data SD1 is [t3, t4], and the period of the temporary test data SD2 is [t7, t8].

[0056] A blank period G1(t2, t3) exists between the temporary training data RD1 and the temporary test data SD1. A blank period G2(t4, t5) exists between the temporary training data RD2 and the temporary test data SD1. A blank period G3(t6, t7) exists between the temporary training data RD2 and the temporary test data SD2. A blank period G4(t8, t9) exists between the temporary training data RD3 and the temporary test data SD2. The four blank periods G1 to G4 do not belong to the periods of the three pieces of temporary training data RD1 to RD3, and do not belong to the periods of the two pieces of temporary test data SD1 and SD2.

[0057] According to the second modification of the first embodiment, the training data generation apparatus 1 divides the operation data OD into a plurality of pieces of temporary test data SD. The training data generation apparatus 1 detects the anomaly value AV from each of the plurality of temporary test data SD based on the plurality of temporary training data RD. Therefore, according to the second modification, the same effects as those of the first embodiment and the first modification can be obtained.

Third Modification of First Embodiment

[0058] FIG. 6 is an illustrative diagram illustrating division processing of the operation data OD according to a third modification of the first embodiment. FIG. 6 sequentially illustrates each processing according to steps S11 and S12 in FIG. 2. A period of each data in each processing is indicated by a band B1D and six patterns P1 to P6.

[0059] As illustrated in a band B1D of a first stage, the acquisition unit 11 acquires the operation data OD (see step S11). The period of the operation data OD is [t1, t7].

[0060] As illustrated in the patterns P1 to P6 in second to seventh stages, the division unit 12 divides the operation data OD into six patterns P1 to P6 related to a combination of the temporary training data RD and the temporary test data SD (see step S12). The division unit 12 may divide the operation data OD such that the period of each of the six pieces of temporary test data SD1 to SD6 is different among the six patterns P1 to P6. The division unit 12 may divide the period [t1, t7] of the operation data OD into the periods of the six pieces of temporary test data SD1 to SD6 in the

six patterns **P1** to **P6**.

[0061] In the pattern **P1**, the period of the temporary test data **SD1** is $[t1, t2]$, and the period of the temporary training data **RD1** is $(t2, t7]$. In the pattern **P2**, the period of the temporary test data **SD2** is $(t2, t3]$, the period of the temporary training data **RD21** is $[t1, t2]$, and the period of the temporary training data **RD22** is $(t3, t7]$. In the pattern **P3**, the period of temporary test data **SD3** is $(t3, t4]$, the period of temporary training data **RD31** is $[t1, t3]$, and the period of temporary training data **RD32** is $(t4, t7]$.

[0062] In the pattern **P4**, the period of the temporary test data **SD4** is $(t4, t5]$, the period of the temporary training data **RD41** is $[t1, t4]$, and the period of the temporary training data **RD42** is $(t5, t7]$. In the pattern **P5**, the period of the temporary test data **SD5** is $(t5, t6]$, the period of the temporary training data **RD51** is $[t1, t5]$, and the period of the temporary training data **RD52** is $(t6, t7]$. In the pattern **P6**, the period of the temporary test data **SD6** is $(t6, t7]$, and the period of the temporary training data **RD6** is $[t1, t6]$. The periods of two adjacent temporary test data **SD** are continuous at a boundary time (start time and end time).

[0063] The detection unit **13** detects the anomaly value **AV** from the temporary test data **SD** based on the temporary training data **RD** in each of the six patterns **P1** to **P6** (see step **S13**). In the pattern **P1**, the detection unit **13** detects the anomaly value **AV** from the temporary test data **SD1** based on the temporary training data **RD1**. In the pattern **P2**, the detection unit **13** detects the anomaly value **AV** from the temporary test data **SD2** based on the temporary training data **RD21** and **RD22**. In the pattern **P3**, the detection unit **13** detects the anomaly value **AV** from the temporary test data **SD3** based on the temporary training data **RD31** and **RD32**.

[0064] In the pattern **P4**, the detection unit **13** detects the anomaly value **AV** from the temporary test data **SD4** based on the temporary training data **RD41** and **RD42**. In the pattern **P5**, the detection unit **13** detects the anomaly value **AV** from the temporary test data **SD5** based on the temporary training data **RD51** and **RD52**. In the pattern **P6**, the detection unit **13** detects the anomaly value **AV** from the temporary test data **SD6** based on the temporary training data **RD6**. In this manner, the detection unit **13** can detect the anomaly value **AV** from the entire period of the operation data **OD** $[t1, t7]$.

[0065] The periods of the six pieces of temporary test data **SD1** to **SD6** do not overlap each other. In another example, the periods of the temporary test data **SD** may overlap each other. In a case where duplication is permitted, the total period of each period of the temporary test data **SD** may correspond to the entire period of the operation data **OD**. In this case, the detection unit **13** can detect the anomaly value **AV** from the entire period of the operation data **OD**.

[0066] The generation unit **14** generates the training data **XD** by excluding the anomaly period in which the anomaly value **AV** is detected in the six patterns **P1** to **P6** from the period of the operation data **OD** $[t1, t7]$ (see step **S14**).

[0067] FIG. 7 is an illustrative diagram illustrating detection processing of an anomaly value **AV** according to a third modification of the first embodiment. FIG. 7 shows a time-series graph **200** relating to the operation data **OD** of FIG. 6. The horizontal axis of the time-series graph **200** indicates time. The vertical axis of the time-series graph **200** indicates the value of the process variable included in the operation data **OD**. The time-series data **210** indicates a temporal change related to the value of the process variable by a solid line and a dotted line. In the time-series data **210**, portions **211** and **212** in the anomaly period in which the anomaly value **AV** is detected are emphasized by dotted lines.

[0068] A period $[t1, t7]$ of the time-series data **210** is divided into periods of six pieces of temporary test data **SD1** to **SD6**. An upper broken line **220** and a lower broken line **230** in the period $[t1, t2]$ of the temporary test data **SD1** correspond to the maximum value and the minimum value of the temporary training data **RD1**, respectively (see pattern **P1**). The upper broken line **220** and the lower broken line **230** in the period $(t2, t3]$ of the temporary test data **SD2** correspond to the maximum value and the minimum value of the temporary training data **RD21** and **RD22**,

respectively (see pattern P2). The upper broken line **220** and the lower broken line **230** in the period (t3, t4] of the temporary test data SD3 correspond to the maximum value and the minimum value of the temporary training data RD31 and RD32, respectively (see pattern P3).

[0069] The upper broken line **220** and the lower broken line **230** in the period (t4, t5] of the temporary test data SD4 correspond to the maximum value and the minimum value of the temporary training data RD41 and RD42, respectively (see pattern P4). The upper broken line **220** and the lower broken line **230** in the period (t5, t6] of the temporary test data SD5 correspond to the maximum value and the minimum value of the temporary training data RD51 and RD52, respectively (see pattern P5). The upper broken line **220** and the lower broken line **230** in the period (t6, t7] of the temporary test data SD6 correspond to the maximum value and the minimum value of the temporary training data RD6, respectively (see pattern P6).

[0070] The time-series graph **200** may be displayed on a display device (see FIG. 20). The user can visually recognize the displayed time-series graph **200** and recognize a temporal change of the time-series data **210**. The user can visually recognize the portions **211** and **212** in the time-series data **210** and easily recognize the anomaly value AV and the anomaly period. The user can recognize that the value of the time-series data **210** exceeds the upper broken line **220** in the period (t4, t5] of the temporary test data SD4 based on the portion **211**. Based on the portion **212**, the user can recognize that the value of the time-series data **210** falls below the lower broken line **230** in the period (t5, t6] of the temporary test data SD5.

[0071] According to the third modification of the first embodiment, the training data generation apparatus **1** divides the operation data OD into a plurality of patterns related to a combination of the temporary training data RD and the temporary test data SD. The training data generation apparatus **1** detects the anomaly value AV from the temporary test data SD based on the temporary training data RD in each of the plurality of patterns. Therefore, the training data generation apparatus **1** can detect the anomaly value AV from a plurality of periods in the operation data OD. Since the training data generation apparatus **1** excludes each anomaly period in which the anomaly value AV is detected from the operation data OD, the training data generation apparatus **1** can generate the training data XD having less anomaly data than the first embodiment.

Fourth Modification of First Embodiment

[0072] FIG. 8 is an illustrative diagram illustrating division processing of the operation data OD according to a fourth modification of the first embodiment. FIG. 8 sequentially illustrates each processing according to steps S11 and S12 of FIG. 2. A period of each data in each processing is indicated by a band B1E and six patterns P1 to P6.

[0073] As illustrated in a band B1E of a first stage, the acquisition unit **11** acquires the operation data OD (see step S11). The period of the operation data OD is [t1, t7].

[0074] As illustrated in the patterns P1 to P6 in second to seventh stages, the division unit **12** divides the operation data OD into six patterns P1 to P6 related to a combination of the temporary training data RD and the temporary test data SD (see step S12). In the six patterns P1 to P6, the periods of the temporary test data SD1 to SD6 are similar to those in the third modification. In particular, the lengths of the periods of the temporary test data SD1 to SD6 are the same.

[0075] In the pattern P1, a blank period G1 exists between the temporary training data RD1 and the temporary test data SD1. In the pattern P2, a blank period G21 exists between the temporary training data RD21 and the temporary test data SD2, and a blank period G22 exists between the temporary training data RD22 and the temporary test data SD2. In the pattern P3, a blank period G31 exists between the temporary training data RD31 and the temporary test data SD3, and a blank period G32 exists between the temporary training data RD32 and the temporary test data SD3.

[0076] In the pattern P4, a blank period G41 exists between the temporary training data RD41 and the temporary test data SD4, and a blank period G42 exists between the temporary training data RD42 and the temporary test data SD4. In the pattern P5, a blank period G51 exists between the temporary training data RD51 and the temporary test data SD5, and a blank period G52 exists

between the temporary training data RD52 and the temporary test data SD5. In the pattern P6, a blank period G6 exists between the temporary training data RD6 and the temporary test data SD6. In particular, a length of each of the ten blank periods G1 to G6 is the same and corresponds to $\frac{1}{2}$ of the length of each period of the six pieces of temporary test data SD1 to SD6.

[0077] The detection unit 13 detects the anomaly value AV from the temporary test data SD based on the temporary training data RD in each of the six patterns P1 to P6 (see step S13). A method of detecting the anomaly value AV is similar to that in the third modification.

[0078] The generation unit 14 generates the training data XD by excluding the anomaly period in which the anomaly value AV is detected in the six patterns P1 to P6 from the period of the operation data OD [t1, t7](see step S14). A method of generating the training data XD is similar to that of the third modification.

[0079] FIG. 9 is a first illustrative diagram illustrating detection processing of the anomaly value AV according to the fourth modification of the first embodiment. FIG. 9 illustrates a time-series graph 300A regarding the operation data OD of FIG. 8. The time-series graph 300A is similar to the time-series graph 200 of FIG. 7. In a time-series data 310, portions 311A and 312A in the anomaly period in which the anomaly value AV is detected are emphasized by dotted lines.

[0080] A period [t1, t7] of the time-series data 310 is divided into a period of six pieces of temporary test data SD1 to SD6. An upper broken line 320 and a lower broken line 330 in the period [t1, t2] of the temporary test data SD1 correspond to the maximum value and the minimum value of the temporary training data RD1, respectively (see pattern P1). The upper broken line 320 and the lower broken line 330 are similar to the upper broken line 220 and the lower broken line 230 in FIG. 7, respectively.

[0081] Unlike the third modification, in the fourth modification, a blank period G exists between the temporary training data RD and the temporary test data SD. That is, the length of the period of the temporary training data RD is different between the third modification and the fourth modification. Therefore, the positions of the upper broken line 320 and the lower broken line 330 indicating the maximum value and the minimum value in the temporary training data RD may be different from the positions of the upper broken line 220 and the lower broken line 230 in FIG. 7.

[0082] The time-series graph 300A may be displayed on a display device (see FIG. 20). The user can visually recognize the time-series graph 300A and recognize a temporal change of the time-series data 310. The user can visually recognize the portions 311A and 312A in the time-series data 310 and easily recognize the anomaly value AV and the anomaly period. In particular, the period (anomaly period) corresponding to the portion 311A is longer than the period of the portion 211 in FIG. 7.

[0083] On the other hand, the period (anomaly period) corresponding to the portion 312A is the same as the period of the portion 212 in FIG. 7.

[0084] FIG. 10 is a second illustrative diagram illustrating the detection processing of the anomaly value AV according to the fourth modification of the first embodiment. A time-series graph 300B of FIG. 10 is a graph in which an upper one-dot chain line 350 and a lower one-dot chain line 360 are superimposed on the time-series graph 300A of FIG. 9. A portion 311B exceeding the upper one-dot chain line 350 in the time-series data 310 indicates a true anomaly value AV. A portion 312B below the lower one-dot chain line 360 of the time-series data 310 indicates a true anomaly value AV. The true anomaly value AV is an ideal anomaly value to be detected from the time-series data 310. The period in which the true anomaly value AV is detected can also be said to be a true anomaly period.

[0085] The time-series graph 200 in FIG. 7 is compared with the time-series graph 300B in FIG. 10. The period (anomaly period) of the portion 211 in the time-series graph 200 is much shorter than the period (true anomaly period) of the portion 311B in the time-series graph 300B and does not substantially coincide with each other. That is, in the third modification according to FIG. 7, the training data generation apparatus 1 detects only a part of the true anomaly period out of the entire

true anomaly period, and does not detect the remaining most of the true anomaly periods.

[0086] The time-series graph **300A** in FIG. **9** is compared with the time-series graph **300B** in FIG. **10**. The period (anomaly period) of the portion **311A** in the time-series graph **300A** is slightly longer than the period (true anomaly period) of the portion **311B** in the time-series graph **300B**, and includes the period of the portion **311B**. That is, in the fourth modification according to FIG. **9**, the training data generation apparatus **1** detects the entire true anomaly period. As compared with the third modification, the training data generation apparatus **1** according to the fourth modification detects the anomaly value (or anomaly period) more appropriately, and thus, can generate the training data XD more appropriately.

[0087] According to the fourth modification of the first embodiment, the training data generation apparatus **1** divides the operation data OD into a plurality of patterns related to a combination of the temporary training data RD and the temporary test data SD. The training data generation apparatus **1** divides the operation data OD such that a blank period G exists between the temporary training data RD and the temporary test data SD in each of the plurality of patterns.

[0088] Therefore, according to the fourth modification, the similar effects to those of the first modification and the third modification can be obtained. In the fourth modification, the anomaly value AV is less likely to be mixed in the training data XD as compared with the third modification. As a result, the training data generation apparatus **1** can generate training data XD having higher quality than that of the third modification.

Second Embodiment

[0089] FIG. **11** is a functional configuration diagram of a training data generation apparatus **1** according to a second embodiment. In the second embodiment, the training data generation apparatus **1** further includes a display control unit **15** and a selection unit **16** in addition to the acquisition unit **11**, the division unit **12**, the detection unit **13**, and the generation unit **14**.

[0090] The display control unit **15** controls various displays. The display control unit **15** receives various data (example: operation data OD, temporary training data RD, temporary test data SD, anomaly value AV) from the acquisition unit **11**, the division unit **12**, and the detection unit **13**. The display control unit **15** generates the image data GD based on the received various data. The display control unit **15** outputs the generated image data GD to the outside of the training data generation apparatus **1** (for example, a display device).

[0091] The selection unit **16** performs various selections. The selection unit **16** generates a selection instruction SI corresponding to the selection operation SO in response to the selection operation SO from the user. The selection unit **16** transmits the generated selection instruction SI to the generation unit **14**. In response to a feedback operation from the user, the selection unit **16** may generate a feedback instruction corresponding to the feedback operation and transmit the feedback instruction to the generation unit **14**.

[0092] FIG. **12** is an operation flowchart of the training data generation apparatus **1** according to the second embodiment. The training data generation apparatus **1** sequentially executes each processing according to steps S21 to S26.

[0093] (Step S21) First, the training data generation apparatus **1** acquires the operation data OD through the acquisition unit **11**. Step S21 is similar to step S11. The acquisition unit **11** transmits the operation data OD to the division unit **12**, the generation unit **14**, and the display control unit **15**.

[0094] (Step S22) Next, the training data generation apparatus **1** divides the operation data OD through the division unit **12**. Step S22 is similar to step S12. The division unit **12** transmits the temporary training data RD and the temporary test data SD to the detection unit **13** and the display control unit **15**.

[0095] (Step S23) Subsequently, the training data generation apparatus **1** detects the anomaly value AV through the detection unit **13**. Step S23 is similar to step S13. The detection unit **13** transmits the anomaly value AV to the generation unit **14** and the display control unit **15**.

[0096] (Step S24) Subsequently, the training data generation apparatus **1** displays an image through the display control unit **15**. The display control unit **15** generates the image data GD based on the operation data OD, the temporary training data RD, the temporary test data SD, and the anomaly value AV obtained in steps S21 to S23. The display control unit **15** displays an image based on the image data GD on the display device.

[0097] The display control unit **15** may display the time-series graph **200** of FIG. 7 or may display the time-series graph **300A** or **300B** of FIG. 9 or **10** on the display device as the image. According to the time-series graph **300A** of FIG. 9, the anomaly value AV and the anomaly period in the time-series data **310** are indicated by portions **311A** and **312A**. Like portions **311A** and **312A**, the anomaly portions may be indicated by different line types (e.g., dotted lines). Alternatively, the anomaly portion may be indicated by a different color. That is, the anomaly portion may be shown in a manner distinguishable from the normal portion. With such a display mode, the user can easily recognize the anomaly portion.

[0098] In a case where the detection unit **13** detects the anomaly value AV for a plurality of process variables included in the operation data OD, the display control unit **15** may display a time-series graph related to each of the plurality of process variables. The display control unit **15** may display each of the time-series graphs in parallel. The display control unit **15** may display a time-series graph related to the process variable having a first ID. The display control unit **15** may switch from the currently displayed time-series graph to another time-series graph for display in accordance with an instruction from the user.

[0099] Conversely, in a case where the detection unit **13** has not detected the anomaly value AV for each of the plurality of process variables included in the operation data OD, the display control unit **15** may display a character string indicating that the anomaly value AV has not been detected.

[0100] (Step S25) Subsequently, the training data generation apparatus **1** selects a process variable through the selection unit **16**. The selection unit **16** receives a selection operation SO (or a feedback operation) from the user for the image displayed in step S24. The selection operation SO may be an operation of selecting a specific process variable that the user has determined to be truly anomaly. The selection operation SO may be an operation in which the user selects a desired anomaly period from among the anomaly periods in the time-series graph of the process variable. The selection operation SO may be input through an input device (see FIG. 20). The selection operation SO may be an operation of selecting a specific process variable determined not to be anomaly by the user, or may be an operation of selecting a specific period determined not to be anomaly by the user. The training data generation apparatus **1** can perform various types of processing (for example, identification of an anomaly period) according to a user operation by determining an item to be selected by the user in advance.

[0101] In a case where the user selects a specific process variable, a check box corresponding to each of the plurality of process variables may be displayed on the display device as a graphical user interface (GUI). In a case where the user selects a desired check box, a selection instruction SI for selecting a process variable related to the check box is generated. The selection unit **16** transmits the selection instruction SI corresponding to the various selection operations SO to the generation unit **14**.

[0102] (Step S26) Finally, the training data generation apparatus **1** generates the training data XD through the generation unit **14**. The generation unit **14** performs processing corresponding to the selection instruction SI transmitted in step S25. In a case where the selection instruction SI is an instruction to select a specific process variable, the generation unit **14** generates the training data XD by excluding the anomaly period from the time-series data corresponding to the specific process variable. In a case where the selection instruction SI is an instruction to select a specific anomaly period, the generation unit **14** generates the training data XD by excluding the specific anomaly period from the time-series data of the process variable.

[0103] In a case where there is no process variable corresponding to the selection instruction SI, the

generation unit **14** generates the operation data OD as the training data XD as it is. In a case where there is a plurality of process variables corresponding to the selection instruction SI, the generation unit **14** excludes each anomaly period detected in the time-series data of each of the plurality of process variables from the operation data OD and generates the training data XD.

[0104] According to the second embodiment, the training data generation apparatus **1** generates the training data XD based on the selection operation SO from the user. The training data generation apparatus **1** generates the training data XD by excluding the anomaly period from the time-series data of the process variable for the process variable determined to be truly anomaly by the user.

Alternatively, for an anomaly period determined by the user to be truly anomaly, the training data generation apparatus **1** generates the training data XD by excluding the anomaly period from the time-series data of the process variable. Therefore, in a case where the training data generation apparatus **1** erroneously detects a period which is originally a normal period as an anomaly period, the user does not select this period, so that it is possible to prevent the period from being excluded.

[0105] Further, the training data generation apparatus **1** displays the anomaly period of the operation data OD in a mode distinguishable from the normal period. With such a display mode, the user can easily recognize the anomaly period of the operation data OD and easily select the anomaly period as the exclusion target. The user automatically generates the training data XD by selecting a process variable indicating an anomaly behavior.

[0106] Therefore, the training data generation apparatus **1** can reduce the labor of the user to manually generate the training data XD.

First Modification of Second Embodiment

[0107] The detection unit **13** may detect the anomaly value AV from the temporary test data SD based on the statistic of the process variable related to the temporary training data RD. The detection unit **13** may detect the value of the temporary test data SD not included in a range from an L percentile to an H percentile of the temporary training data RD as the anomaly value AV. For example, L=5 and H=95. In the case of L=0, the 0 percentile corresponds to the minimum value. In the case of H=100, the 100 percentile corresponds to the maximum value. In the case of L=0 and H=100, the first modification corresponds to the second embodiment.

[0108] For example, L>0 and H<100 are satisfied. In this case, the training data generation apparatus **1** can easily detect the true anomaly value AV.

Second Modification of Second Embodiment

[0109] The detection unit **13** may detect a value whose frequency (or normalized frequency) is less than a threshold in a frequency distribution (or normalized frequency distribution) of the process variable related to the temporary training data RD as the anomaly value AV from the temporary test data SD. The detection unit **13** detects a value present in the temporary training data RD at a frequency less than the threshold as the anomaly value AV in the temporary test data SD.

[0110] FIG. **13** is a frequency distribution diagram **400A** of the temporary training data RD according to a second modification of the second embodiment. FIGS. **13(A)** to **13(F)** illustrate the normalized frequency distributions of the temporary training data RD**1** to RD**6** corresponding to the temporary test data SD**1** to SD**6**, respectively, in order from the left. The frequency distribution diagram **400A** corresponds to the patterns P**1** to P**6** of FIG. **8** and may be displayed on the display device.

[0111] As illustrated in FIGS. **13(A)** to **13(F)**, solid lines indicate the normalized frequencies of the temporary training data RD**1** to RD**6**. The broken line indicates a threshold for the normalized frequency. In FIG. **13(A)**, in the temporary test data SD**1**, a value larger than the value VA**1** and a value smaller than the value VA**2** are detected as the anomaly values AV. In FIG. **13(B)**, in the temporary test data SD**2**, a value larger than the value VB**1** and a value smaller than the value VB**2** are detected as the anomaly values AV. In FIG. **13(C)**, in the temporary test data SD**3**, a value larger than the value VC**1** and a value smaller than the value VC**2** are detected as the anomaly values AV. In FIG. **13(D)**, in the temporary test data SD**4**, a value larger than the value VD**1** and a value

smaller than the value VD2 are detected as the anomaly values AV. In FIG. 13(E), in the temporary test data SD5, a value larger than the value VE1 and a value smaller than the value VE2 are detected as the anomaly values AV. In FIG. 13(F), in the temporary test data SD6, a value larger than the value VF1 and a value smaller than the value VF2 are detected as the anomaly values AV. [0112] According to the second modification of the second embodiment, the similar effects to those of the second embodiment can be obtained.

Third Modification of Second Embodiment

[0113] The display control unit 15 may display the frequency distribution (or normalized frequency distribution) of the process variable related to the temporary training data RD. Furthermore, the display control unit 15 may display the frequency distribution (or normalized frequency distribution) of the process variable related to the temporary test data SD for the temporary test data SD corresponding to the temporary training data RD.

[0114] FIG. 14 is a frequency distribution diagram 400B according to a third modification of the second embodiment. FIGS. 14(A) to 14(F) illustrate normalized frequency distributions of the temporary training data RD1 to RD6 and the temporary test data SD1 to SD6 in order from the left. The frequency distribution diagram 400B corresponds to the patterns P1 to P6 of FIG. 8 and may be displayed on the display device.

[0115] As illustrated in FIGS. 14(A) to 14(F), the solid line indicates the normalized frequency distribution (first distribution) of the temporary training data RD. A one-dot chain line indicates a normalized frequency distribution (second distribution) of the temporary test data SD. HI (Histogram Intersection) indicates the similarity between the first distribution and the second distribution. Instead of the similarity, another index (examples: Kullback-Leibler Divergence, Jensen-Shannon Divergence, Wasserstein Metric) may be used. A first distribution and a second distribution may be distinguishably displayed by different display colors or the like. At least one of the first distribution, the second distribution, and the similarity between the first distribution and the second distribution may be displayed.

[0116] The larger the value of HI, the more similar the first distribution and the second distribution. For example, in a case where the value of HI is less than the threshold, it is defined that the temporary test data SD is not similar to the temporary training data RD. In a case where the threshold is 0.3, the distributions illustrated in FIGS. 14 (C) to 14(E) satisfy $HI < 0.3$. Therefore, the training data generation apparatus 1 determines that the temporary test data SD3, SD4, and SD5 are not similar to the corresponding training data RD3, RD4, and RD5, respectively.

[0117] In a case where HI is less than the threshold, HI may be color-coded, or HI may be surrounded by a frame line. Alternatively, a frame line may be added to the distribution in which HI is less than the threshold, or the distribution may be colored with different colors. In this manner, the training data generation apparatus 1 can highlight and display the temporary test data SD including the anomaly value AV. In a case where the distribution degree of difference is used, the training data generation apparatus 1 may determine that the anomaly value AV is included in the temporary test data SD in a case where the distribution degree of difference exceeds the threshold.

[0118] According to the third modification of the second embodiment, the training data generation apparatus 1 displays the distribution of the temporary test data SD and the distribution of the temporary training data RD.

[0119] The user can easily recognize the difference in distribution between the data. Further, the training data generation apparatus 1 displays a similarity (for example, HI) between the distributions. The user can quantitatively evaluate the similarity of the distribution between the two and easily select a process variable or an anomaly period that is truly determined to be anomaly.

Fourth Modification of Second Embodiment

[0120] The detection unit 13 may detect the anomaly value AV from the temporary test data SD using the machine training model trained on the temporary training data RD (see step S23). As the machine training model, a known method may be used (see Patent Literature 2: Japanese Patent

Publication No. 2022-074890). Hereinafter, the temporary training data RD and the temporary test data SD of FIG. 8 are used.

[0121] The detection unit **13** trains a machine training model for the temporary test data SD1 using the temporary training data RD1 (see pattern P1). The detection unit **13** trains a machine training model for the temporary test data SD2 using the temporary training data RD21 and RD22 (see pattern P2). The detection unit **13** trains a machine training model for the temporary test data SD3 using the temporary training data RD31 and RD32 (see pattern P3).

[0122] The detection unit **13** trains a machine training model for the temporary test data SD4 using the temporary training data RD41 and RD42 (see pattern P4). The detection unit **13** trains a machine training model for the temporary test data SD5 using the temporary training data RD51 and RD52 (see pattern P5). The detection unit **13** trains a machine training model for the temporary test data SD6 by the temporary training data RD6 (see pattern P6). In this manner, the detection unit **13** trains a total of six machine training models in the six patterns P1 to P6.

[0123] The detection unit **13** uses the machine training model trained for the temporary test data SD1 to generate a predicted value in the period [t1, t2] of the temporary test data SD1. Similarly, the detection unit **13** generates the predicted value in the period of the temporary test data SD2 to SD6 using each machine training model trained for the temporary test data SD2 to SD6. The detection unit **13** transmits the generated predicted value to the display control unit **15**.

[0124] The detection unit **13** calculates the deviation by subtracting the predicted value corresponding to the temporary test data SD from the value (measured value) of the process variable included in temporary test data SD. The detection unit **13** transmits the calculated deviation to the display control unit **15**.

[0125] The detection unit **13** detects a timing at which the deviation exceeds the upper threshold and a timing at which the deviation falls below the lower threshold. The detection unit **13** detects the value of the process variable at both timings as the anomaly value AV. The detection unit **13** transmits the anomaly value AV to the display control unit **15**.

[0126] FIG. 15 is an illustrative diagram illustrating anomaly value detection processing according to a fourth modification of the second embodiment. A time-series graph 300C of FIG. 15(A) is similar to the time-series graph 300A of FIG. 9. A time-series graph 500 in FIG. 15(B) shows the deviation.

[0127] As illustrated in FIG. 15(A), a time-series data 340 indicating the predicted value is superimposed on the time-series graph 300C. The predicted value is a value predicted by the six machine training models described above. The time-series data 340 indicates a temporal change in the predicted value by a one-dot chain line. The value of the time-series data 310 greatly deviates from the value of the time-series data 340 in the period corresponding to the portions 311A and 312A (in particular, the portion 311A).

[0128] As illustrated in FIG. 15(B), the time-series graph 500 illustrates a deviation between the time-series data 310 and the time-series data 340 in the time-series graph 300C. The horizontal axis of the time-series graph 500 indicates time. The vertical axis of the time-series graph 500 indicates the deviation. The time-series data 510 indicates a temporal change in the deviation by a dotted line and a solid line. An upper broken line 520 indicates an upper threshold of the deviation. A lower broken line 530 indicates a lower threshold of the deviation.

[0129] The dotted line portion of the time-series data 510 indicates that the deviation is included in a value range between the upper threshold and the lower threshold. A solid line portion of the time-series data 510 indicates that the deviation deviates from the value range (that is, the anomaly value AV). A portion 511 of the time-series data 510 indicates that the deviation is above the upper threshold. A portion 512 of the time-series data 510 indicates that the deviation is below the lower threshold. The portions 511 and 512 correspond to portions 311A and 312A, respectively. The portions 511 and 512 may be displayed in a mode distinguishable from the normal portion by different display colors, line thicknesses, and the like. Such a display mode is user-friendly because

the user can easily recognize the anomaly portion.

[0130] According to the fourth modification of the second embodiment, the training data generation apparatus **1** calculates the predicted value for the value (measured value) of the temporary test data SD using the machine training model trained on the temporary training data RD. The training data generation apparatus **1** displays a time-series graph relating to the deviation between the measured value and the predicted value. Therefore, the training data generation apparatus **1** can present the anomaly value AV and the anomaly period in the operation data OD, which can be easily recognized by the user.

Fifth Modification of Second Embodiment

[0131] In the time-series graph **500** of deviation illustrated in FIG. **15(B)**, the user may determine that the portion **511** of the time-series data **510** is anomaly and the portion **512** is not anomaly. In this case, the user may perform a selection operation SO for selecting the portion **511**. The selection unit **16** may generate the selection instruction SI corresponding to the selection operation SO and transmit the generated selection instruction SI to the generation unit **14**. The generation unit **14** generates the training data XD by excluding the anomaly period related to the selection instruction SI from the operation data OD.

[0132] The selection unit **16** may receive feedback indicating in which period of the process variable the anomaly is recognized. In this case, the display control unit **15** may display a graphical user interface (GUI) (example: text box, calendar) for inputting a start time and an end time related to the period. The user may specify, through the displayed GUI, a period during which the user determines that the user is anomaly among the period of the portion **511** and the period of the portion **512**. The user can specify the time period in more detail through the displayed GUI.

[0133] The display control unit **15** may display a button for adding or deleting a GUI for the user to input the anomaly period. Accordingly, even in a case where one process variable includes a plurality of anomaly periods, the user can specify the plurality of anomaly periods.

Third Embodiment

[0134] FIG. **16** is a functional configuration diagram of a training data generation apparatus **1** according to a third embodiment. As in the second embodiment, the training data generation apparatus **1** includes an acquisition unit **11**, a division unit **12**, a detection unit **13**, a generation unit **14**, a display control unit **15**, and a selection unit **16**.

[0135] The acquisition unit **11** further acquires reference data FD in addition to operation data OD. The reference data FD is data having a period different from that of the operation data OD. The operation data OD and the reference data FD may be data relating to the operation state of the same device and in different periods. A relationship between the operation data OD and the reference data FD is similar to the relationship between the temporary test data SD and the temporary training data RD. The acquisition unit **11** transmits the operation data OD and the reference data FD to the detection unit **13**.

[0136] FIG. **17** is an operation flowchart of the training data generation apparatus **1** according to the third embodiment. The training data generation apparatus **1** sequentially executes each processing according to steps S31 to S38.

[0137] (Steps S31 to S33) First, the training data generation apparatus **1** acquires the operation data OD through the acquisition unit **11**. Next, the training data generation apparatus **1** divides the operation data OD through the division unit **12**. Subsequently, the training data generation apparatus **1** detects the anomaly value AV through the detection unit **13**. Steps S31 to S33 are similar to steps S21 to S23, respectively.

[0138] (Step S34) Subsequently, the training data generation apparatus **1** acquires the reference data FD through the acquisition unit **11**.

[0139] (Step S35) Subsequently, the training data generation apparatus **1** detects the anomaly value AV through the detection unit **13**. The detection unit **13** assumes that the reference data FD acquired in step S34 is data in a normal state (normal data), and detects an anomaly value AV

(another anomaly value AV) from the operation data OD. As a method of detecting the anomaly value AV, the above-described arbitrary method can be applied.

[0140] The detection unit **13** may determine whether or not a value of the operation data OD is included in a value range between the maximum value and the minimum value related to the reference data FD. The detection unit **13** may calculate the predicted value of the operation data OD using the machine training model trained on the reference data FD. The detection unit **13** may determine whether or not the deviation between the predicted value and the value (measured value) of the operation data OD is included in the value range between the upper threshold and the lower threshold.

[0141] (Steps S36 to S38) Subsequently, the training data generation apparatus **1** displays an image through the display control unit **15**. Subsequently, the training data generation apparatus **1** selects a process variable through the selection unit **16**. Finally, the training data generation apparatus **1** generates the training data XD through the generation unit **14**. Steps S36 to S38 are similar to steps S24 to S26, respectively.

[0142] FIG. **18** is an illustrative diagram illustrating anomaly value AV detection processing according to the third embodiment. A time-series graph **600** in FIG. **18(A)** is similar to the time-series graph **300** C in FIG. **15(A)**. A time-series graph **700** in FIG. **18(B)** is similar to the time-series graph **500** in FIG. **15(B)**.

[0143] As illustrated in FIG. **18(A)**, the time-series graph **600** is a graph illustrating a temporal change in the operation data OD. A time-series data **610** indicates a temporal change regarding the value of the process variable of the operation data OD by a solid line and a dotted line. An upper broken line **620** indicates a maximum value of the reference data FD. A lower broken line **630** indicates a minimum value of the reference data FD. Time-series data **640** indicates a predicted value based on the machine training model trained on the reference data FD by a one-dot chain line. A portion **611** of the time-series data **610** indicates a portion (that is, the anomaly value AV) exceeding the upper broken line **620** in the time-series data **610**. A portion **612** of the time-series data **610** indicates a portion (that is, the anomaly value AV) of the time-series data **610** below the lower broken line **630**.

[0144] The user can visually recognize the portions **611** and **612** and easily recognize the anomaly value AV in the time-series data **610**. The portions **611** and **612** may be distinguishably displayed by display modes such as different colors and different line thicknesses.

[0145] As illustrated in FIG. **18(B)**, the time-series graph **700** illustrates a deviation between the value (measured value) of the process variable of the operation data OD and the predicted value. The time-series data **710** indicates a temporal change in the deviation by a dotted line and a solid line. An upper broken line **720** indicates an upper threshold for the deviation. A lower broken line **730** indicates a lower threshold for the deviation. A portion **711** of the time-series data **710** indicates a portion (that is, the anomaly value AV) exceeding the upper broken line **720** in the time-series data **710**. A portion **712** of the time-series data **710** indicates a portion (that is, the anomaly value AV) of the time-series data **710** below the lower broken line **730**.

[0146] The user can visually recognize the portions **711** and **712** and easily recognize the anomaly value AV in the time-series data **710**. The portions **711** and **712** may be distinguishably displayed by display modes such as different colors and different line thicknesses.

[0147] FIG. **19** is a frequency distribution diagram **800** of the operation data OD and the reference data FD according to the third embodiment. The frequency distribution diagram **800** of FIG. **19** is similar to the frequency distribution diagram **400B** of FIG. **14**.

[0148] In the frequency distribution diagram **800**, a one-dot chain line indicates a normalized frequency distribution of the operation data OD. A solid line indicates a normalized frequency distribution of the reference data FD. The similarity between the two distributions is displayed as “HI=0.5”.

[0149] In a case where there is a transition (shift) between the distribution of the operation data OD

and the reference data FD by visually recognizing the frequency distribution diagram **800**, the user can easily recognize the transition. The frequency distribution may be displayed instead of the normalized frequency distribution. Instead of HI, another index (described above) indicating the degree of difference between the two distributions may be displayed.

[0150] FIG. **20** is a hardware configuration diagram of the training data generation apparatus **1** according to each embodiment. The training data generation apparatus **1** includes a CPU **111**, a RAM **112**, a ROM **113**, a storage **114**, a display device **115**, an input device **116**, and a communication device **117** as components. The components are communicably connected to each other by an internal bus. The training data generation apparatus **1** may include at least some of the components.

[0151] The CPU **111** is a processor that executes various processes in accordance with a program. The CPU **111** uses a predetermined area of the RAM **112** as a work area. The CPU **111** reads and executes each program stored in the ROM **113** or the storage **114** to implement each unit (example: acquisition unit **11**, division unit **12**, detection unit **13**, generation unit **14**, display control unit **15**, selection unit **16**). The CPU **111** is an example of a processing unit.

[0152] The RAM **112** is a memory that rewritably stores various data. For example, the RAM **112** is a synchronous dynamic random access memory (SDRAM). The RAM **112** is an example of a storage unit.

[0153] The ROM **113** is a memory that stores various data in a non-rewritable manner. The ROM **113** is an example of a storage unit.

[0154] The storage **114** is various storage media. The storage **114** may be a drive apparatus that writes or reads various data in or from a storage medium. The storage **114** may write or read various data to or from a storage medium according to control by the CPU **111**. The storage **114** is an example of a storage unit.

[0155] The display device **115** is an apparatus that displays various data. The display device **115** may be a liquid crystal display (LCD). The display device **115** displays various data (examples: image, time-series graph, distribution diagram) based on a display signal from the CPU **111**. The display device **115** is an example of a display unit.

[0156] The input device **116** is an apparatus that receives various input operations from the user. The input device **116** may be a mouse or a keyboard. The input device **116** receives an operation input by the user as an instruction signal, and transmits the instruction signal to the CPU **111**. The input device **116** is an example of an input unit.

[0157] The communication device **117** communicates with an external device through a network in accordance with control by the CPU **111**. The communication device **117** is an example of a communication unit.

[0158] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions.

[0159] The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A training data generation apparatus comprising a processor configured to: acquire operation data related to an operation state of a device in a predetermined period; divide the operation data into at least temporary training data and temporary test data; detect an anomaly value from the temporary test data based on the temporary training data; and generate training data from the operation data by excluding an anomaly period in which the anomaly value is detected from the

predetermined period.

2. The apparatus according to claim 1, wherein the processor is further configured to divide the operation data such that a blank period exists between the temporary training data and the temporary test data.

3. The apparatus according to claim 1, wherein the processor is further configured to: divide the operation data into a plurality of patterns related to a combination of the temporary training data and the temporary test data, and detect the anomaly value from the temporary test data based on the temporary training data in each of the patterns.

4. The apparatus according to claim 3, wherein the processor is further configured to divide the operation data such that each period of the temporary test data differs among the patterns.

5. The apparatus according to claim 3, wherein the processor is further configured to divide the predetermined period of the operation data into each period of the temporary test data in the patterns.

6. The apparatus according to claim 3, wherein the processor is further configured to divide the operation data such that a blank period exists between the temporary training data and the temporary test data in each of the patterns.

7. The apparatus according to claim 1, wherein the processor is further configured to: display a time-series graph related to the operation data on a display device, and emphasize a portion of the anomaly period in the time-series graph.

8. The apparatus according to claim 7, wherein the processor is further configured to: select at least one anomaly period of each of the anomaly period in the time-series graph in response to an operation from a user, and generate the training data by excluding the selected anomaly period from the predetermined period.

9. The apparatus according to claim 1, wherein the operation data includes time-series data related to each of a plurality of process variables, and the processor is further configured to: display a time-series graph related to the time-series data on a display device, and emphasize a portion of the anomaly period in the time-series graph.

10. The apparatus according to claim 9, wherein the processor is further configured to: select at least one process variable of each of the process variables in response to an operation from a user, and generate the training data by excluding the anomaly period from the predetermined period for the time-series data corresponding to the selected process variable.

11. The apparatus according to claim 1, wherein the processor is further configured to display at least one of a first distribution related to the temporary training data, a second distribution related to the temporary test data, and a similarity between the first distribution and the second distribution on a display device.

12. The apparatus according to claim 1, wherein the processor is further configured to detect the anomaly value from the temporary test data using a machine training model trained on the temporary training data.

13. The apparatus according to claim 1, wherein the processor is further configured to: acquire reference data having a period different from a period of the operation data, detect another anomaly value from the operation data based on the reference data, and generate the training data by excluding another anomaly period in which the another anomaly value is detected from the predetermined period.

14. A training data generation method causing a computer to perform: acquiring operation data related to an operation state of a device in a predetermined period; dividing the operation data into at least temporary training data and temporary test data; detecting an anomaly value from the temporary test data based on the temporary training data; and generating training data from the operation data by excluding an anomaly period in which the anomaly value is detected from the predetermined period.

15. A non-transitory computer readable medium including computer executable instructions,

wherein the instructions, when executed by a processor, cause the processor to perform a method comprising: acquiring operation data related to an operation state of a device in a predetermined period; dividing the operation data into at least temporary training data and temporary test data; detecting an anomaly value from the temporary test data based on the temporary training data; and generating training data from the operation data by excluding an anomaly period in which the anomaly value is detected from the predetermined period.
